

Lecture / Week 15Numerical Solution of 1st Order ODEs

1) Euler's Method $y_{n+1} = y_n + h f(x_n, y_n)$

2) Improved Euler Method

$$y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^*)]$$

$$y_{n+1}^* = y_n + h f(x_n, y_n)$$

3) 2nd Order RK-Method.

$$y_{n+1} = y_n + h \left[\frac{k_1}{2} + \frac{k_2}{2} \right]$$

$$k_1 = f(x_n, y_n) \quad k_2 = f(x_{n+1}, y_n + h k_1) \\ = f(x_{n+1}, y_n + h f(x_n, y_n))$$

4) 4th Order RK-Method.

$$y_{n+1} = y_n + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$k_1 = h f(x_n, y_n)$$

$$k_2 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_{n+1}, y_n + k_3)$$

(2)

Example $\frac{dy}{dx} = y - x^2$; $y(0) = 1$

Use 4th order RK-Method with $h=0.1$ to approximate $y(0.5)$

Solution $f(x, y) = y - x^2$ $x_0 = 0$; $y_0 = 1$; $h = 0.1$
 $0 \leq x \leq 0.5$

Formula $y_{n+1} = y_n + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$

$$k_1 = h f(x_n, y_n)$$

$$k_2 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_{n+1}, y_n + k_3)$$

③

n	x_n	y_n	k_1	$x_{n+\frac{h}{2}}$	$y_{n+\frac{k_1}{2}}$	k_2	$y_{n+\frac{k_2}{2}}$	k_3
0	$x_0 = 0$	$y_0 = 1$	$k_1 = hf(x_0, y_0)$ $= 0.1f(0, 1)$ $= 0.1$	$x_0 + \frac{0.1}{2} = \frac{0.1}{2}$ $= 0.05$	$y_0 + \frac{0.1}{2} = 1.05$	$k_2 = hf(0.05, 1.05)$ $= 0.10475$	$y_0 + \frac{k_2}{2}$ $= 1 + \frac{0.10475}{2}$ $= 1.05238$	$k_3 = hf(0.05, 1.05238)$ $= 0.104988$

1	$x_1 = 0.1$	$y_1 = 1.104829$	$k_1 = hf(x_1, y_1)$ $= 0.109483$	$x_1 + \frac{0.1}{2} = 0.1 + \frac{0.1}{2}$ $= 0.15$	$y_1 + \frac{k_1}{2} = 1.15957$	$k_2 = hf$ $= 0.113707$	$y_1 + \frac{k_2}{2} =$	$k_3 = 0.113918$
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$$y_{n+1} = y_n + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$y_1 = y(0.1) = y_0 + \frac{1}{6} [0.1 + 2(0.10475) + 2(0.104988) + 0.109499]$$

$$= 1.104829$$

$$y_2 = y(0.2) = 1.104829 + \frac{1}{6} [0.109483 + 2(0.113707) + 2(0.113918) + 0.117875]$$

$$= 1.218597$$

x_{n+1}	y_{n+k_3}	k_4
$x_1 = 0.1$	$y_0 + k_3$ $= 1 + (0.104988)$ $= 1.104988$	$k_4 = hf(0.1, 1.104988)$ $= 0.109499$

$x_2 = 0.2$	$y_1 + k_3$ $=$	$k_4 = 0.117875$
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Numerical Solution of Higher Order ODEs

(4)

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_1 \frac{dy}{dx} + a_0 y = f(x)$$

where a 's ~~are~~ are the coefficients

→ If a 's are constants, the ODE is called higher order ODE with constant coefficients

→ If a 's are function of 'x', the ODE is called higher order ODE with variable coefficients.

→ If a 's are function of 'y', the ODE is called ~~higher order ODE with~~ non-linear higher order ODE.

We only consider higher order ODEs with constant coefficients.

If $f(x)$ is zero it is called higher order homogeneous ODE.

If $f(x) \neq 0$ then it is called non-homogeneous ODE.

Initial and Boundary Conditions.

⑤

Explain with an example

Let $3\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = e^{-x}$

$$\begin{array}{l|l} y(0) = 5 & y(0) = 5 \\ y'(0) = 7 & y(2) = 6 \end{array}$$

Initial Conditions

Boundary Conditions.

In our course we only consider higher ODEs with Initial conditions

Example

$$\frac{d^3y}{dx^3} - 4\frac{d^2y}{dx^2} - y = x^2$$

$$y(1) = 1 ; y'(1) = -2 ; y''(1) = 2.$$

Numerical Solution of higher Order ODEs using Euler's Method.

⑥

Example $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = e^{-x}$

$$y(0) = 1 \quad y'(0) = 2.$$

Use Euler's method with $h = 0.75$ to approximate $y(0.75)$

Solution

~~Step~~
Step I :- Let $y = z \rightsquigarrow$ ①

$$\frac{dy}{dx} = \frac{dz}{dx} = u \rightsquigarrow$$
 ②

$$\frac{d^2 y}{dx^2} = \frac{d^2 z}{dx^2} = \frac{du}{dx} \rightsquigarrow$$
 ③

Substitute (1)-(3) in the given ODE.

$$\frac{du}{dx} + 2u + z = e^{-x} \rightsquigarrow$$
 (A)

$$\frac{dz}{dx} = u \rightsquigarrow$$
 (B)

From I.C.

$$z(0) = 1 \quad u(0) = 2$$

we have

$$\frac{du}{dx} = e^{-x} - 2u - z; \quad u(0) = 2 \quad \left| \quad \frac{dz}{dx} = u; \quad z(0) = 1 \right.$$

Example 2

(7)

$$3 \frac{d^3 y}{dx^3} - 2 \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} - y = x^2 \rightarrow (1)$$

$$y(1) = 1 ; y'(1) = -2 ; y''(1) = 4$$

Solution

Divide Eq. (1) with 3.

$$\frac{d^3 y}{dx^3} - \frac{2}{3} \frac{d^2 y}{dx^2} + \frac{4}{3} \frac{dy}{dx} - \frac{1}{3} y = \frac{x^2}{3} \rightarrow (2)$$

$$\text{Let } y = z \rightarrow (3)$$

$$\frac{dy}{dx} = \frac{dz}{dx} = u \rightarrow (4)$$

$$\frac{d^2 y}{dx^2} = \frac{d^2 z}{dx^2} = \frac{du}{dx} = v \rightarrow (5)$$

$$\frac{d^3 y}{dx^3} = \frac{d^3 z}{dx^3} = \frac{d^2 u}{dx^2} = \frac{dv}{dx} \rightarrow (6)$$

Use (3)-(6) into (2)

$$\frac{dv}{dx} - \frac{2}{3} v + \frac{4}{3} u - \frac{1}{3} z = \frac{x^2}{3}$$

$$\frac{du}{dx} = v$$

$$\frac{dz}{dx} = u$$

$$\text{I.C.s } z(1) = 1 ; u(1) = -2 ; v(1) = 4$$

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$$\frac{dv}{dx} = \frac{2}{3}v - \frac{4}{3}u + \frac{1}{3}z + \frac{x^2}{2} ; v(1) = 4$$

$$\frac{du}{dx} = v ; u(1) = -2$$

$$\frac{dz}{dx} = u ; z(1) = 1$$

Example ③

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = e^{-x}$$

$$h = 0.25$$

$$y(0) = 1$$

$$y'(0) = 2$$

$$y(0.75) = ?$$

Solution

$$\frac{du}{dx} = e^{-x} - 2u - z \rightarrow f_2(x, z, u)$$

$$u(0) = 2$$

$$\frac{dz}{dx} = u \rightarrow f_1(x, z, u)$$

$$z(0) = 1$$

Euler's Method

$$z_{n+1} = z_n + h f_1(x_n, z_n, u_n) = z_n + h (u_n)$$

$$u_{n+1} = u_n + h f_2(x_n, z_n, u_n) = u_n + h (e^{-x_n} - 2u_n - z_n)$$

n	x_n	z_n	u_n	$f_1(x_n, z_n, u_n)$	$f_2(x_n, z_n, u_n)$
0	$x_0 = 0$	$z_0 = 1$	$u_0 = 2$	$f_1(x_0, z_0, u_0) = f(0, 1, 2) = 2$	$f_2(x_0, z_0, u_0) = f(0, 1, 2) = -4$
1	$x_1 = 0.25$	$z_1 = 1.5$	$u_1 = 1$	$f_1(x_1, z_1, u_1) = f(0.25, 1.5, 1) = 1$	$f_2(x_1, z_1, u_1) = f(0.25, 1.5, 1) = -2.7211$
2	$x_2 = 0.5$	$z_2 = 1.75$	$u_2 = 0.31970$	$f_1(x_2, z_2, u_2) = 0.31970$	$f_2(x_2, z_2, u_2) = -1.7289$

$$u_{n+1} = u_n + h f_2(x_n, z_n, u_n)$$

$$u_1 = u_0 + h f_2(x_0, z_0, u_0) = 2 + (0.25)(-4) = 1$$

$$u_2 = u_1 + h f_2(x_1, z_1, u_1) = 1 + (0.25)(-2.7211) = 0.31970$$

$$u_3 = u_2 + h f_2(x_2, z_2, u_2) = 0.31970 + (0.25)(-1.7289) = -0.117289$$

$$z_{n+1} = z_n + h f_1(x_n, z_n, u_n)$$

$$z_1 = y_1 = y(0.25) = z_0 + h f_1(x_0, z_0, u_0) = 1 + (0.25)(2) = 1.5$$

$$z_2 = y_2 = y(0.5) = z_1 + h f_1(x_1, z_1, u_1) = 1.5 + (0.25)(1) = 1.75$$

$$z_3 = y_3 = y(0.75) = z_2 + h f_1(x_2, z_2, u_2) = 1.75 + (0.25)(0.31970) = 1.8299$$

Ans.